Spectral–Spatial Synergy Guided Attention Network for Hyperspectral Image Classification

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***Abstract*—The attention mechanism has gained significant popularity in hyperspectral image classification (HSIC) for its ability to adaptively highlight key features. However, it often incurs expensive computational costs due to the iterative operations. Moreover, the high interclass similarity and indistinct class boundaries in hyperspectral images (HSIs) pose challenges for classification tasks. To tackle these issues, we propose a spectral–spatial synergy guided attention network (SSSGAN) for HSIC. It initially employs a spectral–spatial synergistic mechanism to jointly extract local spectral and spatial features of HSIs, and then applies a guided attention mechanism to effectively reduce computational complexity and further improve its discriminative capabilities. Specifically, a stacked residual spectral feature extractor (SRSFE) stacks scale-variable convolution blocks within the channel-connected residual (CCR) structure. It precisely localizes boundaries and reduces boundary blur by extracting and fusing local multilevel spectral features. A hierarchical dual feature enhancement (HDFE) module employs parallel multi-scale branches (MSs) combined with dual shift operations to extract multiscale spatially enhanced features, thereby enriching the diversity of feature representations and improving the discriminability of interclass features. Finally, a feature-aware guided attention (FAGA) module reduces the computational cost of the attention mechanism by incorporating the guiding token and convolutional operations. By introducing local regularization into the attention map, the convolution enhances the robustness and effectiveness of global and contextual information extraction from HSIs. Experiments conducted on four publicly available benchmark datasets (PU, HU2013, SV, and LK) demonstrate that the proposed SSSGAN method achieves overall accuracies of 98.96%, 95.93%, 98.72%, and 98.31%, respectively, thereby validating its effectiveness in HISC tasks. The source code for this work can be found at https://github.com/szq0816/DSGAN**

***Index Terms*—Attention mechanism, convolutional neural networks (CNNs), dual shift network (DSN), guiding tokens, hyperspectral image classification (HSIC), local and global features.**

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I. Introduction

HYPERSPECTRAL images (HSIs), as a high-dimensional, multispectral information source, provide more comprehensive spectral information compared to traditional color images. They encompass a wide range of spectral features while exhibiting distinct spatial distribution properties. Therefore, HSIs hold significant potential for practical applications, such as urban planning [1], environmental monitoring [2], mineral exploration [3], and disaster monitoring [4]. HSI classification (HSIC) technology aims to accurately distinguish land cover types and has been applied in various practical applications over the past decades [5], [6].

In recent years, many convolutional neural networks (CNNs)-based HSIC methods have been developed and achieved superior classification performances. Hyperspectral features can be effectively extracted using CNNs with different dimensional configurations. Specifically, 1-D CNNs focus on modeling local spectral dependencies at the pixel level [7], while 2-D CNNs exploit spatial contextual information to capture local spatial structures [8]. To leverage the intrinsic 3-D nature of HSIs, 3-D CNNs jointly extract spectral and spatial features by processing the full spectral–spatial cube, thereby enhancing the representational capacity [9]. Furthermore, hybrid 2-D–3-D convolutional architectures offer a more comprehensive modeling of spectral–spatial characteristics [10], although they typically result in increased model complexity and computational overhead, which may lead to overfitting in scenarios with limited training samples. To mitigate these issues, residual frameworks have been introduced to preserve information flow and stabilize network training. For example, spectral–spatial residual networks (SSRN) employ residual connections to alleviate performance degradation [11], and multiscale residual blocks have been designed to enhance feature diversity and reduce overfitting through lightweight, efficient architectures [12]. Although CNN-based HSIC methods show promising performance in local feature extraction, they still struggle to preserve details of irregular interclass boundaries and small-scale irregular regions. Furthermore, they frequently fail to leverage contextual information from local neighborhoods, resulting in the loss of fine-grained features in HSIs.

In recent years, vision transformer (ViT) has made remarkable progress in computer vision owing to its strong global

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modeling capabilities, facilitating its widespread adoption in diverse scenarios [13]. The attention mechanism, as a core component of the ViT architecture [14], can effectively capture long-range dependencies within the input data. It enhances the modeling of global context information and focuses on key information relevant to classification while disregarding irrelevant content. In recent years, several works have applied the transformer architecture to HSIC tasks. These include, for example, cross attention [15], [16] or adaptive relevance estimation strategy [17] to explore the dependencies between local and neighborhood regions.

Although the transformer architecture excels at capturing global features, its ability to model local features is relatively limited. Thus, relying solely on the transformer for HSIC often fails to achieve superior classification performance. To solve this issue, many HSIC methods combine CNNs and transformer architectures to extract the local and global features of HSIs. To explore both shallow multiscale spectral and deep local spatial information, a hybrid hierarchical multigranularity window shifting transformer network was proposed [18], integrating dual-branch convolution blocks and a multigranularity window attention module to enhance spectral–spatial fusion and the discriminative ability of spatial features. To address the limitations of traditional ViTs in modeling short-range spectral features, a long-short distance spatial–spectral cross-attention mechanism was introduced [19]. Additionally, a dual-cross symmetric attention mechanism was designed to enhance feature interaction and fusion after multiscale mapping, enabling the capture of broader dependencies [20]. Although these methods significantly improve classification performance, they still face high computational and storage overhead, and challenges such as interclass similarity and indistinct class boundaries hinder the extraction of highly discriminative features.

To address these issues, we propose a novel spectral–spatial synergy guided attention network (SSSGAN) method for HSIC. First, we design a stacked residual spectral feature extractor (SRSFE) and a hierarchical dual feature enhancement (HDFE) module to effectively capture the local multilevel spectral and multiscale spatially enhanced features of HSI patches. These extracted spectral and spatial features are fused and then fed into the feature-aware guided attention (FAGA) module. In this module, we perform global modeling for the fused local spectral–spatial features. Finally, we decompose a self-attention (SA) operation into two cascaded sub-operations and introduce the guiding token as their respective query and key, thereby efficiently reducing the computational cost. Extensive experiments on four benchmark datasets show the effectiveness of the proposed SSSGAN method in the HSIC task.

The main contributions of this work are summarized as follows.

1) The proposed SSSGAN model employs a spectral–spatial synergy architecture comprising the SRSFE and HDFE modules, which not only collaboratively extract local spectral and spatial features but also enhance the discriminability of interclass features. Specifically, the SRSFE module utilizes a serial architecture of convolutional blocks with varying receptive fields, where shallow layers with small kernels capture fine-grained spectral details, and deeper layers with larger kernels integrate these features into broader contexts, thereby reducing class boundary ambiguity and improving interclass separability. The HDFE module enhances interclass distinction through dual shift operations, mitigating the adverse effects of high interclass similarity on classification performance. Additionally, its parallel multi-scale branches (MSs) extract diverse features across various scales and spatial hierarchies, further improving interclass discriminability.
2) The proposed FAGA module introduces the guiding token to facilitate more efficient information aggregation, significantly reducing computational overhead. Additionally, it integrates convolutional operations into attention to regularize attention distribution, thereby enhancing the structural consistency and robustness of feature representations across local and global contexts.
3) Comprehensive experiments conducted on four benchmark HSI datasets demonstrate that our proposed approach outperforms several state-of-the-art methods in real-world HSIC tasks.

The structure of this article is organized as follows. Section II provides an overview of related work. Section III introduces the architecture of the proposed SSSGAN method. Section IV presents the experimental results, and Section V concludes the work.

II. Related Work

In this section, we provide an overview of HSIC methods based on CNNs and the attention mechanism.

A. CNN-Based HSIC Methods

With the widespread application of CNNs, numerous efforts have been made to apply them to HSIC tasks. Due to the 3-D structure and high spectral dimensionality inherent in HSIs, Roy et al. [10] proposed a hybrid 2-D–3-D convolutional method to jointly extract spectral–spatial features without excessive computational complexity. However, the extensive use of 2-D and 3-D convolution operations leads to high computational costs. To alleviate this problem, Cihan and Ceylan [21] adopted convolution kernels based on spectral characteristics tailored to the data, and then selectively applied them to relevant pixels, significantly reducing the number of model parameters. Ma et al. [22] proposed a lightweight hybrid convolutional neural network, which not only greatly reduces the computational overhead but also achieves excellent classification performance with limited training data. Ma et al. [23] introduced an ultralightweight hybrid CNN model based on redundancy removal, significantly reducing computational resource consumption and enabling more efficient classification tasks. To address the challenges of limited computational and storage resources, Li et al. [24] proposed a channel-level lightweight network, achieving excellent classification performance with fewer channels and layers. Moreover, it

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shows significant advantages in parameter count, model complexity, and inference time. However, most existing methods adopt fixed-size feature extraction, overlooking the interaction between spectral and spatial dimensions. To address this issue, Wang et al. [25] proposed a multiscale fusion network based on global weighting, which integrates spectral and spatial-related information through a 3-D weighting mechanism and reduces both homogeneous and heterogeneous interference, thereby achieving efficient classification with fewer training samples. However, HSIs exhibit highly similar spectral features across different classes, accompanied by ambiguous class boundaries. Traditional CNN-based HSIC models often struggle to effectively distinguish subtle spectral variations due to this interclass similarity, leading to suboptimal classification accuracy.

B. Attention-Based HSIC Methods

The attention mechanism mimics human visual focus [26], allowing models to concentrate on the most relevant portions of the data while disregarding less important information. Due to its outstanding global modeling capabilities, several attention-based HSIC methods have been proposed over the past few years. To effectively learn highly discriminative features from HSIs, Lu et al. [27] proposed a novel band grouping-guided multiattention network for the HSIC problem. It explores significant intragroup information within multiple adaptively divided, nonoverlapping spectral band groups, and then applies attention mechanisms to suppress redundant information, thus extracting group-specific spectral–spatial features. However, these methods often fail to effectively focus on the synergistic relationship of spectral and spatial features, leading to insufficient information extraction. To tackle this problem, Shang et al. [28] introduced a pyramid coordinate attention mechanism for HSIC tasks. It aggregates spatial and local–global feature information and then dynamically enhances the spectral features of the extracted spatial components, enabling a joint focus on spectral–spatial features. Li et al. [29] proposed a novel compressed feature multihead SA module that efficiently compresses spectral–spatial features while maintaining excellent feature extraction capability. Moreover, it significantly improves computational efficiency compared to traditional SA mechanisms. Paoletti et al. [30] introduced a parameter-free attention network, which directly calculates the attention score of the feature map based on the linear separability of neurons, thus achieving higher classification accuracy with fewer parameters. Although the attention mechanism significantly improves the classification performance of HSIs, it also introduces several challenges, including high computational costs and noise sensitivity.

III. Method

In this section, we provide a detailed introduction to the overall framework of the proposed SSSGAN method.

A. Overview of Proposed SSSGAN Method

As illustrated in Fig. 1, the proposed framework consists of three main components: the SRSFE module, the HDFE module, and the FAGA module. First, the original HSI is processed through a 3-D convolutional layer to generate fixed-size HSI patches $X \in R^{p\times p\times s\times c}$, where p represents the height and width of these patches, s denotes the number of spectral channels, and c denotes the channel dimension. Next, the SRSFE and HDFE modules are used to extract local multilevel spectral features and multiscale spatially enhanced features from X, respectively, which are then fused through channel connections and convolution operations. Subsequently, the fused features are input into the FAGA module to model global dependencies while enhancing local structural coherence via attention regularization. Finally, the extracted feature map is fed into global average pooling and then passed through a linear layer to generate the final classification results.

B. Stacked Residual Spectral Feature Extractor

To address the boundary-blurring issue in HSIs, we design the SRSFE module. By integrating the semantic information of high-level features with the detailed information of low-level features through the stacked channel-connected residual (CCR) structure, it can more accurately localize boundaries and alleviate boundary blur. Specifically, the SRSFE module consists of four stacked convolution blocks with identical structures, where each convolution block contains a 3-D convolution layer, a batch normalization (BN) layer, and a Sigmoid activation function. These convolution blocks are sequentially connected in increasing kernel size: $1\times1\times3$, $1\times1\times7$, $1\times1\times11$, and $1\times1\times15$, respectively. The selected kernel sizes are designed to capture both local and global spectral correlations. Smaller kernels (e.g., 3 and 7) focus on fine-grained spectral patterns, whereas larger kernels (e.g., 11 and 15) model broader interband dependencies, thereby enhancing the module's ability to discriminate subtle spectral variations. Let the input of each convolution block be denoted as Z^i_{in}, and its output Z^i_{out} can be expressed as follows:

$$\mathbf{Z}^i_{\text{out}} = \text{Sigmoid}\left(\text{BN}\left(\text{Conv3D}\left(\mathbf{Z}^i_{\text{in}}\right)\right)\right),\ i = 1,2,3,4 \tag{1}$$

where i denotes the ith convolution block, Conv3D$(\cdot)$ is the 3-D convolutional operation, and Sigmoid$(\cdot)$ is the sigmoid activation function. In previous HSIC methods, residual connections were employed to mitigate the gradient vanishing problem, thereby improving the model's generalization capability. Additionally, several approaches utilized channel connections to enhance feature diversity and integrate multiscale information. Thus, the proposed SRSFE module combines them to extract local spectral features. Specifically, the inputs to the second, third, and fourth convolution blocks are derived from channel connections between the outputs of the previous convolution blocks and the original input X. Thus, we have

$$\mathbf{Z}^i_{\text{in}} = \text{Concat}\left(\mathbf{Z}^{i-1}_{\text{out}}, \mathbf{X}\right),\ i = 2,3,4 \tag{2}$$

where Concat$(\cdot,\cdot)$ represents the concatenation operation along the channel dimension. The SRSFE module employs a serial stacking strategy instead of parallel MSs to enable progressive fusion of spectral features. This structure allows deeper blocks to refine and contextualize the outputs of the previous layers,

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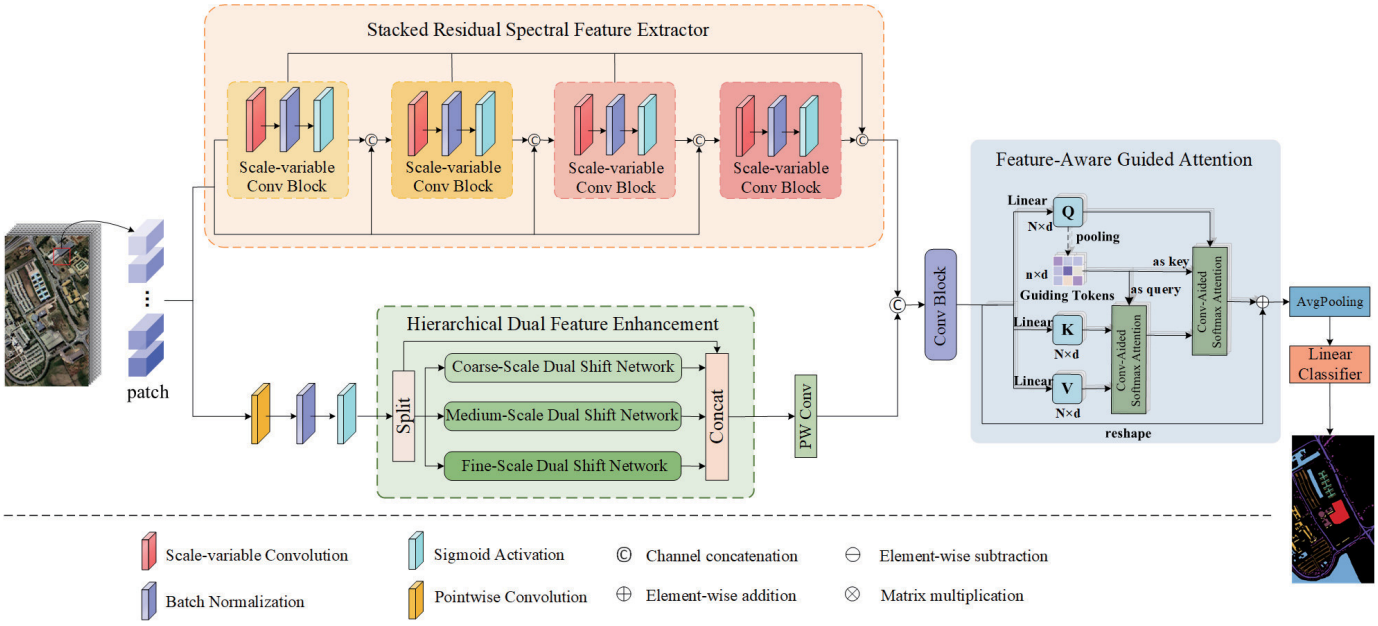


Fig. 1. Overall architecture of the proposed SSSGAN method. It consists of three modules: the SRSFE module, the HDFE module, and the FAGA module. Specifically, the SRSFE module stacks four scale-variable convolution blocks. The HDFE module hierarchically uses three parallel branches to extract spatial enhancement features from coarse to fine. The FAGA module combines guiding tokens and convolutions to enhance the robustness of the proposed model.

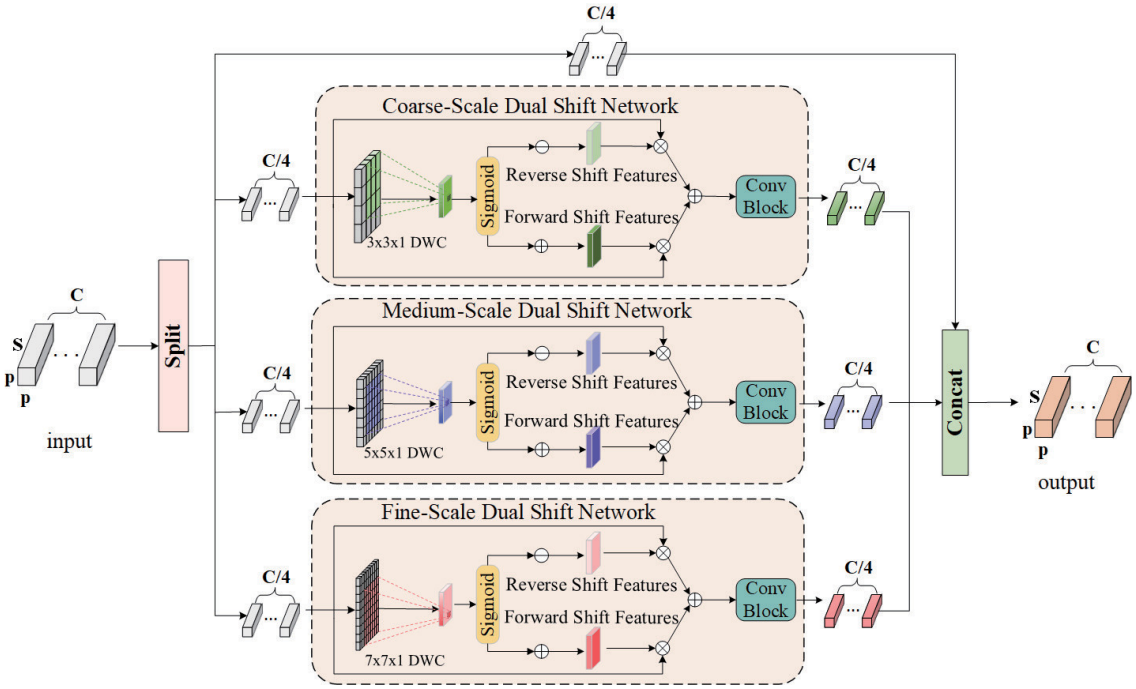


Fig. 2. Structure of the HDFE module.

thereby enhancing boundary localization and improving spectral discriminability. Finally, the outputs of each convolution block are fused through channel connections to generate the output Z_F of the SRSFE module. The process can be expressed as follows:

$$\mathbf{Z}_F = \text{Concat}\left(\mathbf{Z}_{\text{out}}^1,\ \mathbf{Z}_{\text{out}}^2,\ \mathbf{Z}_{\text{out}}^3,\ \mathbf{Z}_{\text{out}}^4\right). \tag{3}$$

C. HDFE Module

To highlight differentiated information in HSIs, we design the HDFE module based on a multiscale dual shift network (DSN). It progressively captures diverse features across multiple spatial scales, thus enhancing the discriminability of class features [31], [32]. As illustrated in Fig. 2, for each small patch $X \in R^{p\times p\times s\times c}$ obtained in the previous operation, we first apply a pointwise convolution to expand its channel dimension to four times its original size and thus obtain the feature map $S \in R^{p\times p\times s\times 4c}$. Subsequently, we divide it along the channel dimension into four sub-blocks: S_1, S_2, S_3, and S_4. The division process can be expressed as follows:

$$\mathbf{S} = \text{PWConv}(\mathbf{X}) \tag{4}$$

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$$\{\mathbf{S}_1, \mathbf{S}_2, \mathbf{S}_3, \mathbf{S}_4\} = \text{Split}(\mathbf{S}) \tag{5}$$

where PWConv(·) is a 3-D pointwise convolution operation and Split(·) represents the division of S along the channel dimension. The convolutional layer is immediately followed by a BN layer and a sigmoid activation function. The sub-blocks S_1, S_2, and S_3 are used as inputs to the multiscale DSN. S_4 serves as a representation of the original features and is fused with the output features processed by the multiscale DSN to generate the final output of the module. The multiscale DSN hierarchically extracts local spatially enhanced features across granularity levels, progressively refining representations from coarse to fine scales. This allows the model to comprehensively explore feature information at different levels, enhancing intercategory discriminability. The HDFE module includes three branches, each employing depth-wise convolution kernels of varying sizes ($3 \times 3 \times 1$, $5 \times 5 \times 1$, and $7 \times 7 \times 1$) to extract feature information. These kernel sizes are tailored to the patch dimensions of specific datasets (e.g., PU: 9×9, LK: 7×7), thus effectively capturing fine, medium, and coarse spatial contexts while ensuring compatibility with the smallest patch sizes. This design also aligns with the physical ground sampling distances, thereby effectively representing structures ranging from small edges to larger regions. The remaining processing steps are identical, enabling hierarchical feature extraction for HSIs. To illustrate the structure in detail, we take one branch of this module as an example. Let $\mathbf{S}_j$ denote the input of the jth branch after channel splitting and depthwise convolution. The intermediate output $\mathbf{Y}_j$ is computed as follows:

$$\mathbf{Y}_j = \text{Sigmoid}\left(\text{DWConv}\left(\mathbf{S}_j\right)\right), \quad j = 1, 2, 3 \tag{6}$$

where j denotes the jth branch and DWConv(·) denotes a 3-D depthwise convolution operation. Subsequently, dual shift operations are applied to simulate reverse and forward spatial shifts, producing the reverse shift feature map $\mathbf{Y}_R^j$ and the forward shift feature map $\mathbf{Y}_F^j$, respectively. The specific operations are defined as follows:

$$\mathbf{Y}_R^j = \mathbf{Y}_j - 1, \quad j = 1, 2, 3 \tag{7}$$

$$\mathbf{Y}_F^j = \mathbf{Y}_j + 1, \quad j = 1, 2, 3. \tag{8}$$

The addition and subtraction operations attempt to distinguish the positive and negative offset directions of the features. This helps emphasize distinctive information and capture subtle differences in the value range of the feature matrix, significantly improving feature extraction performance. Next, the shifted feature maps are used to weight the original input patch $\mathbf{S}_j$ through element-wise multiplication, and the two results are fused by element-wise addition

$$\mathbf{Y}_{\text{add}}^j = \left(\mathbf{Y}_R^j \otimes \mathbf{S}_j\right) \oplus \left(\mathbf{Y}_F^j \otimes \mathbf{S}_j\right), \quad j = 1, 2, 3 \tag{9}$$

where $\otimes$ denotes matrix multiplication and $\oplus$ represents element-wise addition. Finally, a convolution block is applied to reshape the feature map, yielding the output Y_{out} of one branch of the proposed HDFE module. The convolution block comprises a pointwise convolution layer, a BN layer, and a ReLU activation function. The processing of the convolution block can be formulated as follows:

$$\mathbf{Y}_{\text{out}}^j = R\left(\text{BN}\left(\text{PWConv}\left(\mathbf{Y}_{\text{add}}^j\right)\right)\right), \quad i = 1, 2, 3 \tag{10}$$

where $R(\cdot)$ represents the ReLU activation function. To fully integrate the output features from the three branches, we perform a channel concatenation operation, followed by the sequential application of the ReLU activation function and pointwise convolution to fuse the features. The fusion process can be described as follows:

$$\mathbf{Y}_F = \text{PWConv}\left(R\left(\text{Concat}\left(\mathbf{Y}_{\text{out}}^1, \mathbf{Y}_{\text{out}}^2, \mathbf{Y}_{\text{out}}^3, \mathbf{S}_4\right)\right)\right) \tag{11}$$

where Y_F represents the final output of the proposed HDFE module.

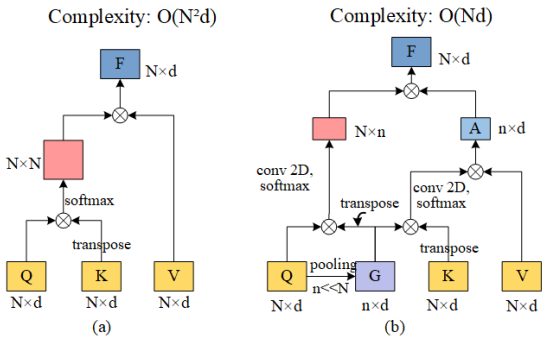


Fig. 3. Visual comparison of computational structure and complexity between traditional (a) SA and (b) proposed FAGA module.

D. FAGA Module

To comprehensively capture global dependencies among features and enhance the model's robustness, we propose the FAGA module in our model. First, we fuse the feature maps Z_F and Y_F obtained from the SRSFE and HDFF modules. Then, the fused features F_{in} are used as the input for the FAGA module. Features F_{in} are then processed through a linear layer to generate the query matrix $Q \in R^{N \times d}$, the key matrix $K \in R^{N \times d}$, and the value matrix $V \in R^{N \times d}$. To reduce the high computational complexity of the SA mechanism while maintaining robust model representation ability, we introduce the guiding token $G \in R^{n \times d}$ in this module, where $n < N$. The guiding token G is generated by pooling the query matrix Q, which can be considered the original query's global representation. The proposed FAGA module reduces computational complexity by replacing full SA with a two-stage guided attention mechanism and integrating convolutional operations to capture local dependencies. This design significantly reduces the attention complexity from $O(n^2 \cdot d)$ to $O(n \cdot d)$, as illustrated in Fig. 3. The FAGA module extends the conventional SA mechanism into two conv-aided softmax attention (CASA) stages. The detailed structure of CASA is illustrated in Fig. 4. In the first stage, the guide token (GT) G serves as the query to aggregate features with the key matrix K and the value matrix V, generating the guided features V_G. In the second stage, the guiding token G serves as the key and the guided features V_G as the value, propagating global information from the guided

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TABLE I
NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR THE PU DATASET

Color	Category	Training	Validation	Testing
	Asphalt	42	42	6547
	Meadows	133	133	18383
	Gravel	14	14	2071
	Trees	21	21	3022
	Metal sheets	14	14	1317
	Soil	35	35	4959
	Bitumen	14	14	1302
	Bricks	28	28	3626
	Shadows	14	14	919
	Total	315	315	42146

features to each query, thereby yielding the output A_{guide}. The process is mathematically expressed as follows:

$$\mathbf{V}_G = \text{CASA}(\mathbf{G}, \mathbf{K}, \mathbf{V}) \tag{12}$$

$$\mathbf{A}_{\text{guide}} = \text{CASA}(\mathbf{Q}, \mathbf{G}, \mathbf{V}_G). \tag{13}$$

Furthermore, to enhance the learning of spectral–spatial joint features and stabilize attention responses, the proposed CASA mechanism incorporates convolution operations into the attention computation as a lightweight regularizer. The specific structure is as follows: First, matrix multiplication is performed between the input query matrix Q and key matrix K to obtain the attention score matrix. This score matrix is then processed through a 2-D convolutional layer and a softmax layer sequentially, and finally multiplied by the value matrix V to generate the aggregated features T_{atten}. Thus, we have

$$\mathbf{T}_{\text{atten}} = \text{Softmax}\left(\text{Conv2D}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{h_d}}\right)\right)\mathbf{V} \tag{14}$$

where h_d denotes the dimension of each attention head, and Conv2D$(\cdot)$ is a 2-D convolutional operation. To improve the model's generalization capability, residual connections are incorporated throughout the FAGA module. Thus, its output A_{out} can be represented as follows:

$$\mathbf{A}_{\text{out}} = \text{Reshape}(\mathbf{F}_{\text{in}}) + \mathbf{A}_{\text{guide}}. \tag{15}$$

The extracted feature map A_{out} is subsequently processed through a global average pooling layer, followed by a linear layer, to produce the final classification outputs.

IV. EXPERIMENTS

In this section, we present experiments to evaluate the effectiveness of the proposed SSSGAN method on four real-world HSI datasets: the Pavia University (PU) dataset, the Houston2013 (HU2013) dataset, the Salinas Valley (SV) dataset, and the WHUHi-LongKou (LK) dataset.

A. HSI Datasets

1) *PU dataset:* The PU dataset was captured using the ROSIS sensor at the University of Pavia, Italy. It spans a spectral range from 0.43 to 0.86 μm and has dimensions of 610 $\times$ 340 pixels. After excluding 12 spectral bands affected by water absorption, 103 spectral bands remained for classification. This dataset includes nine distinct land cover categories as shown in Table I.

TABLE II
NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR THE HU2013 DATASET

Color	Category	Training	Validation	Testing
	Healthy Grass	30	30	1191
	Stressed Grass	30	30	1194
	Synthetic Grass	30	30	637
	Trees	30	30	1184
	Soil	30	30	1182
	Water	30	30	265
	Residential	30	30	1208
	Commercial	30	30	1184
	Road	30	30	1192
	Highway	30	30	1167
	Railway	30	30	1175
	Parking Lot1	30	30	1173
	Parking Lot2	30	30	409
	Tennis Court	30	30	368
	Running Track	30	30	600
	Total	450	450	14129

TABLE III
NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR THE SV DATASET

Color	Category	Training	Validation	Testing
	Weeds_1	40	40	1926
	Weeds_2	60	60	3606
	Fallow	40	40	1894
	Fallow_rough_plow	20	20	1352
	Fallow_smooth	50	50	2577
	Stubble	60	60	3836
	Celery	60	60	3455
	Grapes_untrained	60	60	11143
	Soil_vinyard_develop	40	40	6119
	Corn_senesced_green_weeds	20	20	3236
	Lettuce_romaine_4wk	10	10	1048
	Lettuce_romaine_5wk	10	10	1907
	Lettuce_romaine_6wk	10	10	895
	Lettuce_romaine_7wk	20	20	1029
	Vinyard_untrained	60	60	7142
	Vinyard_vertical_trellis	20	20	1767
	Total	580	580	52932

2) *HU2013 dataset:* The HU2013 dataset was acquired in Houston, Texas, using the ITRES CASI-1500 sensor. It contains 144 spectral bands spanning wavelengths from 0.38 to 1.05 μm. The HSIs in this dataset have a size of 349 $\times$ 1905 pixels, with a spatial resolution of 2.5 m. This dataset consists of 15 land cover categories as shown in Table II.
3) *SV dataset:* The SV dataset was collected using the AVIRIS sensor in the SV, CA, USA. This dataset includes 224 spectral bands spanning wavelengths from 400 to 2500 nm, with an HSI data dimension of 512 $\times$ 217 pixels. After removing water-affected spectral bands, 204 spectral bands were used for classification tasks. The class distribution of the dataset is predominantly agricultural, containing 16 land cover categories as shown in Table III.
4) *LK dataset:* The LK dataset was captured in Longkou Town, Hubei, China, using a DJI M600 Pro drone equipped with an 8 mm focal length imaging sensor. The spatial resolution of the images is approximately

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TABLE IV
NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR THE LK DATASET

Color	Category	Training	Validation	Testing
	Corn	15	15	34481
	Cotton	10	10	8354
	Sesame	10	10	3011
	Broad-leaf soybean	15	15	63182
	Narrow-leaf soybean	10	10	4131
	Rice	10	10	11834
	Water	15	15	67026
	Roads and houses	12	12	7100
	Mixed weed	13	13	5203
	Total	110	110	204322

0.463 m, with dimensions of 550 × 400 × 270, where 550 and 400 correspond to the pixel counts for the height and width, and 270 denotes the number of spectral bands. The dataset includes nine land cover categories as shown in Table IV.

B. Experimental Setting

All experiments were conducted on a computational platform with an Intel Core i9-14900K CPU, 24 GB of RAM, and an NVIDIA GeForce RTX 4090 GPU, using Python 3.8.20 and Torch 2.4.1.

In our experiments, the original HSIs from the four datasets were partitioned into patches of fixed spatial sizes: 9 × 9 for the PU dataset, 9 × 9 for the HU2013 dataset, 13 × 13 for the SV dataset, and 7 × 7 for the LK dataset. To minimize initialization biases, each experiment was repeated ten times with 300 iterations per trial, and the results were reported as the mean ± standard deviation. The proportions of training samples for the four datasets were set to 0.7%, 3%, 1%, and 0.05%, respectively. Tables I–IV summarize the quantities of training, validation, and testing sets for the PU, HU2013, SV, and LK datasets. For all datasets, the learning rate was set to 0.005, with batch sizes of 64, 32, 64, and 32, and the number of attention heads set to 8, 8, 6, and 8, respectively. These hyperparameter values were set based on preliminary experiments to train the proposed model. To evaluate the model's classification performance, we employed overall accuracy (OA), average accuracy (AA), and Kappa coefficient (Kappa) as metrics.

C. Parameter Analysis

In this subsection, we analyzed the impact of the following hyperparameters on classification performance: patch size, batch size, learning rate, and the number of heads. We varied one hyperparameter by fixing the others at their optimal values. Specifically, we first analyzed the effect of patch size, with all other hyperparameters set to their default values: a batch size of 64, a learning rate of 0.005, and seven attention heads. In subsequent analyses of other hyperparameters, previously studied hyperparameters were fixed at their optimal values for each dataset. The experimental results for each hyperparameter are presented as follows.

1) *Patch size:* In this experiment, we evaluated the impact of the patch size on classification performance. In general, larger patch sizes can enhance the richness of feature information, but may also introduce noises, potentially degrading the classification accuracy. Here, we fixed all other hyperparameters and set the patch size to 5, 7, 9, 11, and 13 for the four datasets. As illustrated in Fig. 5, the classification results for the four datasets initially increase and then decrease as the patch size grows. These experimental results show that the optimal patch sizes for the proposed method on the four HSI datasets were set to 9, 9, 13, and 7, respectively.
2) *Batch size:* The batch size is another crucial factor influencing both training efficiency and model performance. Smaller batch sizes may slow down training and introduce instability, but they require less memory and their noisy gradients can help the model avoid local optima, thus enhancing generalization. Conversely, larger batch sizes not only accelerate model convergence but also increase the risk of overfitting. To determine the optimal batch size for each dataset, we started with a smaller size and experimented with values 16, 32, 64, and 128. From the results in Fig. 6, the optimal batch sizes for the four datasets were empirically set to 64, 32, 64, and 32, respectively.
3) *Learning rate:* To determine the optimal learning rate for the four datasets, we set its value to 0.001, 0.005, 0.01, and 0.05, respectively. Fig. 7 shows the results of the

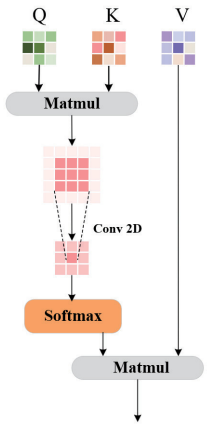


Fig. 4. Structure of the CASA.

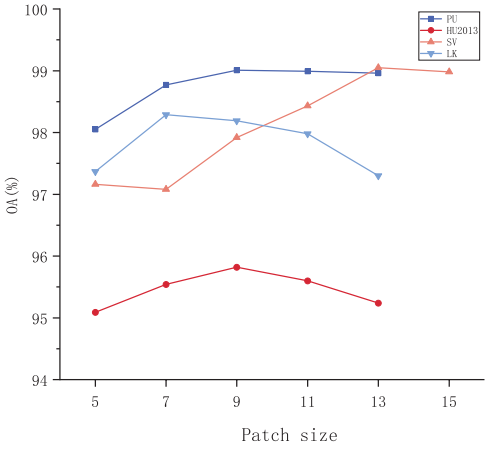


Fig. 5. OA comparison at different patch sizes.

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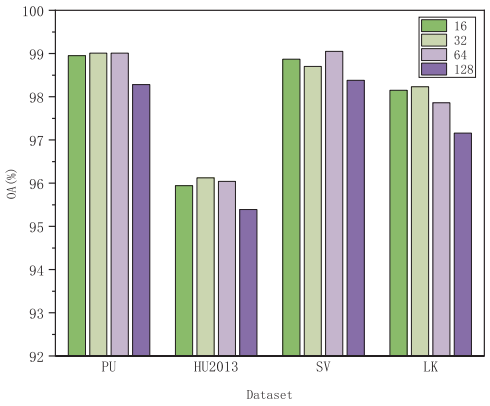


Fig. 6. OA comparison at different batch sizes.

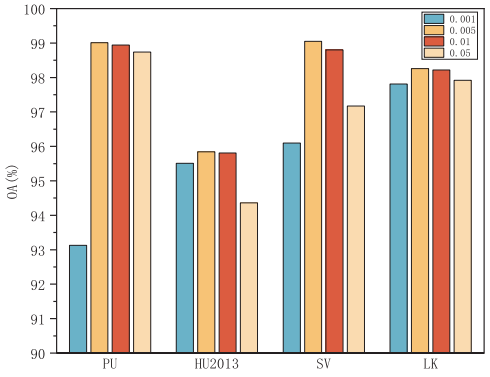


Fig. 7. OA comparison at different learning rates.

proposed method with different learning rate settings. It can be seen that the optimal classification performance across all datasets can be achieved by setting the learning rate to 0.005. Additionally, we can see that a small learning rate may result in slow convergence and potential local optima, while a large rate accelerates convergence but may cause training instability or gradient explosion. To mitigate this problem, we aim to balance the convergence speed and stability by tuning the learning rate.

4) *Number of heads:* The number of heads in the attention mechanism directly affects the model's representation ability and computational efficiency. Increasing the number of heads may increase resource consumption and the risk of overfitting, but it allows the model to capture a broader range of features and dependencies. Hence, we empirically varied the number of heads to evaluate its impact on classification performance. Fig. 8 shows the classification performance of the proposed method varying the number of heads on different HSI datasets. Our method performs optimally with eight heads on the PU, HU2013, and LK datasets, and achieves the best results with six heads on the SV dataset.

D. Comparison Experiments

To validate the superiority of our proposed SSSGAN method, we conducted comprehensive comparisons with ten state-of-the-art HSIC methods, such as SSRN [11], HybridFormer [33], EATN [34], DBMGNet [35], CVSSN [36], DBCTNet [37], FourierFormer [38], DCTN [39], SQSFormer [40], and MassFormer [41]. To ensure a fair comparison, all methods employed the same hyperparameter configurations in classification experiments.

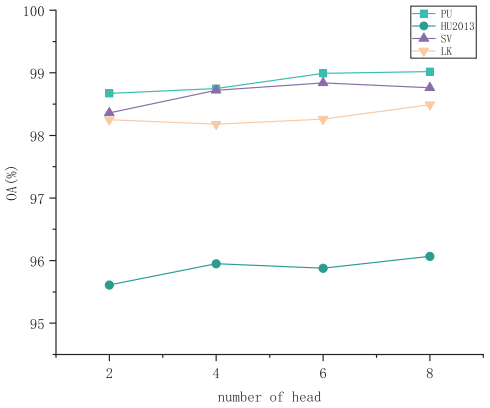


Fig. 8. OA comparison for different head numbers.

1) *Quantitative analysis:* Tables V–VIII present the OA, AA, and Kappa values for various methods on the PU, HU2013, SV, and LK datasets. The best results for each metric are highlighted in bold. Each experiment was repeated ten times to ensure reliability, with results reported as mean ± standard deviation. It can be seen that our proposed method consistently outperforms all other models across all four datasets. Specifically, the overall classification accuracy (OA) for the PU dataset reaches 98.96%, while the HU2013 dataset achieves a significant improvement of 95.93%. The classification accuracies for the SV and LK datasets are 98.72% and 98.31%, respectively. Furthermore, compared to the ten other state-of-the-art HSIC methods, the proposed SSSGAN method not only achieves higher classification accuracy but also shows a lower standard deviation, demonstrating its superiority and stability.

As shown in Table V, the proposed SSSGAN method achieves an OA value of 98.96% on the PU dataset, with a standard deviation lower than that of most comparative methods. Among the nine comparative methods, the HybridFormer model delivers the best performance. This is mainly because this model utilizes a local–global–local strategy, which comprehensively models both spatial and spectral information of HSIs, effectively leveraging features to enhance classification performance. Nevertheless, our SSSGAN method still outperforms it on this dataset, achieving the best classification performance. This is because the proposed SSSGAN model not only models local and global information but also merges multilevel features through stacked scale-variable convolution blocks, which optimize class boundaries and mitigate boundary ambiguity. Furthermore, it employs a multiscale DSN to extract discriminative features at different spatial levels from coarse to fine, significantly enhancing the distinction between interclass features and improving classification accuracy.

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TABLE V
COMPARATIVE EXPERIMENTAL RESULTS ON THE PU DATASET

Class	SSRN [11]	HybridFormer [33]	EATN [34]	DBMGNet [35]	CVSSN [36]	DBCTNet [37]	FourierFormer [38]	DCTN [39]	SQSFormer [40]	MassFormer [41]	DSGAN
1	97.71 ± 0.83	98.66 ± 0.36	95.95 ± 0.75	99.01 ± 0.42	97.39 ± 0.78	97.27 ± 0.86	96.81 ± 0.51	96.67 ± 0.55	95.66 ± 0.63	95.44 ± 1.53	**99.15 ± 0.69**
2	98.86 ± 0.36	99.49 ± 0.13	99.60 ± 0.18	99.44 ± 0.18	99.11 ± 0.41	99.01 ± 0.39	98.39 ± 0.50	96.97 ± 0.70	98.26 ± 0.27	95.52 ± 0.67	**99.78 ± 0.16**
3	96.89 ± 0.98	**97.36 ± 0.82**	95.26 ± 1.28	97.27 ± 0.99	94.27 ± 2.16	91.98 ± 5.65	94.78 ± 2.57	88.74 ± 1.85	89.29 ± 2.33	93.14 ± 2.49	96.15 ± 2.28
4	98.95 ± 0.25	**99.80 ± 0.19**	97.81 ± 0.42	97.68 ± 0.32	97.87 ± 0.78	98.28 ± 0.46	98.54 ± 0.46	98.86 ± 0.28	98.33 ± 0.43	96.79 ± 1.12	98.73 ± 0.33
5	99.81 ± 0.16	**99.89 ± 0.04**	99.83 ± 0.09	99.67 ± 0.08	99.89 ± 0.07	99.75 ± 0.12	99.74 ± 0.36	98.84 ± 0.48	99.82 ± 0.21	96.68 ± 3.04	99.63 ± 0.17
6	97.73 ± 0.75	98.65 ± 0.54	99.14 ± 0.37	99.24 ± 0.40	99.24 ± 0.25	98.61 ± 0.56	98.57 ± 0.57	98.73 ± 0.26	97.50 ± 0.47	88.15 ± 2.09	**99.39 ± 0.39**
7	97.58 ± 2.13	94.44 ± 2.00	96.45 ± 1.65	93.84 ± 2.21	96.10 ± 2.92	98.64 ± 1.72	90.57 ± 2.99	91.00 ± 3.97	90.21 ± 2.71	83.28 ± 3.07	**99.98 ± 0.03**
8	**96.42 ± 0.94**	96.35 ± 0.44	96.26 ± 0.92	95.41 ± 1.58	95.40 ± 1.45	90.59 ± 4.34	90.99 ± 2.45	88.32 ± 1.68	84.92 ± 1.94	93.21 ± 0.89	95.33 ± 1.19
9	99.36 ± 0.55	99.59 ± 0.43	97.45 ± 0.25	96.68 ± 0.28	98.89 ± 1.17	98.17 ± 1.04	99.29 ± 1.01	97.62 ± 0.60	**99.91 ± 0.10**	98.26 ± 1.04	98.07 ± 1.07
OA	98.25 ± 0.22	98.76 ± 0.14	98.22 ± 0.22	98.53 ± 0.18	98.13 ± 0.22	97.53 ± 0.55	97.15 ± 0.36	95.97 ± 0.51	95.94 ± 0.35	94.09 ± 0.49	**98.96 ± 0.17**
AA	98.15 ± 0.28	98.25 ± 0.24	97.53 ± 0.30	97.58 ± 0.32	97.57 ± 0.32	96.92 ± 0.69	96.41 ± 0.61	95.08 ± 0.65	94.88 ± 0.54	93.39 ± 0.78	**98.47 ± 0.20**
Kappa	97.67 ± 0.30	98.36 ± 0.18	97.64 ± 0.29	98.05 ± 0.23	97.52 ± 0.29	96.71 ± 0.73	96.21 ± 0.48	94.63 ± 0.68	94.60 ± 0.47	92.16 ± 0.65	**98.62 ± 0.22**

TABLE VI
COMPARATIVE EXPERIMENTAL RESULTS ON THE HU2013 DATASET

Class	SSRN [11]	HybridFormer [33]	EATN [34]	DBMGNet [35]	CVSSN [36]	DBCTNet [37]	FourierFormer [38]	DCTN [39]	SQSFormer [40]	MassFormer [41]	DSGAN
1	95.31 ± 2.11	94.82 ± 0.66	94.48 ± 0.93	94.43 ± 0.31	94.13 ± 1.29	97.17 ± 2.24	93.92 ± 1.13	94.92 ± 0.88	93.96 ± 0.44	94.54 ± 1.39	**98.81 ± 1.28**
2	97.08 ± 1.70	97.28 ± 0.93	**97.61 ± 0.84**	97.12 ± 0.67	97.61 ± 1.26	96.90 ± 2.05	95.02 ± 1.23	92.72 ± 1.86	95.02 ± 1.22	91.22 ± 5.56	96.48 ± 1.52
3	99.94 ± 0.13	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	99.98 ± 0.05	**100.00 ± 0.00**	99.21 ± 0.69	99.71 ± 0.87	99.92 ± 0.25	98.76 ± 1.47	**100.00 ± 0.00**
4	98.76 ± 0.63	99.24 ± 0.19	98.23 ± 0.87	98.41 ± 0.60	98.34 ± 0.89	97.05 ± 1.20	98.65 ± 0.51	**99.58 ± 0.19**	96.22 ± 1.38	97.60 ± 1.62	97.84 ± 0.86
5	97.79 ± 1.02	97.48 ± 0.59	97.60 ± 0.94	97.47 ± 0.84	97.76 ± 0.82	97.67 ± 1.17	95.68 ± 0.70	94.65 ± 1.24	97.98 ± 0.70	94.92 ± 1.33	**98.05 ± 1.54**
6	95.90 ± 5.22	97.89 ± 1.24	96.83 ± 3.47	94.23 ± 3.78	94.10 ± 3.64	95.61 ± 3.97	77.59 ± 4.25	89.10 ± 5.97	94.36 ± 1.93	93.27 ± 3.96	**99.62 ± 0.50**
7	96.85 ± 0.73	95.84 ± 0.65	94.49 ± 0.47	96.62 ± 0.63	96.50 ± 0.47	95.25 ± 0.89	93.62 ± 1.27	94.25 ± 1.26	93.39 ± 0.90	90.80 ± 3.05	**97.82 ± 0.60**
8	94.10 ± 2.06	**97.02 ± 1.08**	96.56 ± 0.63	95.91 ± 0.86	92.46 ± 1.89	94.73 ± 1.49	93.21 ± 1.54	96.72 ± 1.05	95.33 ± 1.19	91.85 ± 2.23	94.75 ± 1.94
9	92.07 ± 2.08	95.31 ± 0.88	93.63 ± 0.81	**95.57 ± 1.21**	95.05 ± 0.77	91.32 ± 2.08	94.69 ± 1.37	92.25 ± 1.77	91.89 ± 1.79	90.27 ± 2.82	90.13 ± 1.20
10	93.03 ± 2.58	84.66 ± 2.14	91.19 ± 1.69	91.03 ± 0.92	**94.40 ± 1.11**	92.91 ± 2.47	91.43 ± 1.08	80.55 ± 2.68	85.63 ± 1.64	92.44 ± 2.03	93.72 ± 3.57
11	91.15 ± 1.89	90.58 ± 1.24	93.08 ± 1.09	94.99 ± 1.80	95.12 ± 0.96	95.20 ± 1.63	87.84 ± 1.79	88.03 ± 1.89	92.93 ± 1.75	88.79 ± 2.19	**96.44 ± 1.88**
12	84.88 ± 1.81	88.77 ± 1.15	90.05 ± 1.05	95.25 ± 0.45	88.34 ± 1.57	93.56 ± 2.97	77.19 ± 1.67	80.18 ± 2.42	91.22 ± 2.40	82.10 ± 2.51	**96.11 ± 1.56**
13	89.27 ± 6.41	**96.26 ± 0.91**	93.04 ± 2.71	88.87 ± 5.18	90.15 ± 2.67	85.25 ± 4.29	91.54 ± 3.27	90.44 ± 2.32	90.85 ± 1.37	75.49 ± 7.12	86.80 ± 4.80
14	94.10 ± 4.27	94.53 ± 2.00	97.87 ± 1.49	96.42 ± 2.13	**99.52 ± 0.71**	94.10 ± 2.98	90.01 ± 3.15	98.72 ± 2.59	98.23 ± 1.45	90.70 ± 4.50	94.33 ± 4.72
15	**98.30 ± 0.73**	96.72 ± 1.39	95.59 ± 0.87	95.90 ± 1.51	95.86 ± 0.84	97.38 ± 1.47	97.52 ± 1.68	94.60 ± 0.62	94.90 ± 0.62	93.67 ± 0.83	98.23 ± 1.75
OA	94.27 ± 0.34	94.44 ± 0.43	94.99 ± 0.20	95.60 ± 0.30	95.14 ± 0.30	95.09 ± 0.44	91.94 ± 0.40	91.59 ± 0.63	93.70 ± 0.42	91.13 ± 0.70	**95.93 ± 0.29**
AA	94.57 ± 0.48	95.09 ± 0.39	95.35 ± 0.36	95.48 ± 0.59	95.29 ± 0.28	94.94 ± 0.56	91.81 ± 0.36	92.43 ± 0.71	94.12 ± 0.32	91.09 ± 0.69	**95.94 ± 0.42**
Kappa	93.80 ± 0.36	93.99 ± 0.46	94.57 ± 0.21	95.24 ± 0.32	94.75 ± 0.32	94.69 ± 0.47	91.28 ± 0.43	90.91 ± 0.69	93.19 ± 0.45	90.41 ± 0.76	**95.60 ± 0.31**

As shown in Table VI, the proposed SSSGAN model achieves superior classification performance, outperforming other state-of-the-art HSIC methods by at least 0.46% in AA. Unlike on other datasets with high accuracy, all competitors perform poorly on the HU2013 dataset. This may be attributed to the distribution characteristics of land categories in the dataset, such as the blurry category boundaries of buildings, roads, and other urban areas, as well as the similar spectral features of certain classes (such as healthy grassland and synthetic grassland), which increase the difficulty of classification. Nevertheless, the proposed SSSGAN model retains class boundary information effectively through the CCR structure and increases the difference between similar spectra via the dual shift operations. Therefore, it achieves the best classification results in land categories, such as healthy grassland, synthetic grassland, and residential areas.

Tables VII and VIII show the classification performances of various HSIC methods on the SV and LK datasets. The results indicate that the proposed SSSGAN method achieves the best performance across two datasets, further validating its superiority in HSIC. Additionally, we can see that the CVSSN method achieves relatively superior classification performances on both datasets. This is because it exploits the potential relationships between center vectors and neighboring vectors through a self-similarity network and efficiently extracts spectral–spatial features of HSIs. However, the proposed SSSGAN model outperforms the CVSSN model on these two datasets, with OA value improving by 0.77% and 0.52%, respectively. This may be because the CVSSN model fails to adequately focus on the global information of HSIs, thus insufficiently exploring the spatial correlations of the features. In addition, our SSSGAN method efficiently extracts local spectral–spatial features via stacked and hierarchical structures, thereby effectively capturing subtle differences. Moreover, compared to traditional attention-based methods, the proposed module retains global modeling capability while introducing local structural bias through convolutional regularization, thereby enhancing feature consistency and improving classification performance.

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TABLE VII
COMPARATIVE EXPERIMENTAL RESULTS ON THE SV DATASET

Class	SSRN [11]	HybridFormer [33]	EATN [34]	DBMGNet [35]	CVSSN [36]	DBCTNet [37]	FourierFormer [38]	DCTN [39]	SQSFormer [40]	MassFormer [41]	DSGAN
1	99.99 ± 0.02	**100.00 ± 0.00**	99.70 ± 0.36	99.98 ± 0.03	96.57 ± 2.87	99.88 ± 0.34	99.83 ± 0.39	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	99.99 ± 0.02
2	99.99 ± 0.01	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	100.00 ± 0.01	99.88 ± 0.20	**100.00 ± 0.00**	99.96 ± 0.05	97.35 ± 1.31	**100.00 ± 0.00**
3	95.19 ± 0.84	97.58 ± 0.87	97.35 ± 0.86	94.00 ± 0.93	**99.20 ± 0.62**	97.07 ± 1.53	94.97 ± 2.52	97.68 ± 0.53	92.89 ± 1.58	92.88 ± 2.23	98.49 ± 1.10
4	97.19 ± 0.60	97.16 ± 0.32	96.79 ± 0.31	95.53 ± 0.64	97.44 ± 1.85	96.96 ± 0.61	96.77 ± 0.93	96.72 ± 0.93	97.84 ± 0.40	96.68 ± 0.96	**98.49 ± 1.15**
5	**100.00 ± 0.00**	99.73 ± 0.25	99.98 ± 0.04	**100.00 ± 0.00**	99.95 ± 0.08	99.99 ± 0.02	99.80 ± 0.29	**100.00 ± 0.00**	99.61 ± 0.29	**100.00 ± 0.00**	99.33 ± 1.34
6	100.00 ± 0.01	99.98 ± 0.05	99.98 ± 0.04	99.99 ± 0.02	**100.00 ± 0.00**	99.99 ± 0.02	99.95 ± 0.05	99.98 ± 0.03	99.99 ± 0.01	99.97 ± 0.07	**100.00 ± 0.00**
7	99.84 ± 0.10	99.86 ± 0.23	99.38 ± 0.30	98.81 ± 0.71	99.73 ± 0.23	99.86 ± 0.13	99.82 ± 0.22	**99.93 ± 0.08**	99.74 ± 0.26	99.16 ± 0.40	99.70 ± 0.27
8	98.01 ± 0.47	95.56 ± 0.43	96.51 ± 0.74	97.44 ± 0.34	97.87 ± 0.58	98.53 ± 0.61	96.69 ± 0.45	92.82 ± 1.78	93.38 ± 0.91	94.55 ± 1.02	**98.61 ± 1.56**
9	99.22 ± 0.33	99.18 ± 0.13	99.53 ± 0.12	99.72 ± 0.13	99.30 ± 0.13	98.77 ± 0.64	99.60 ± 0.30	99.31 ± 0.12	**99.84 ± 0.13**	99.48 ± 0.33	99.21 ± 0.37
10	99.40 ± 0.59	94.14 ± 0.70	98.82 ± 0.65	97.62 ± 0.59	99.30 ± 0.73	98.80 ± 1.27	99.27 ± 0.76	97.37 ± 1.09	95.50 ± 1.05	94.88 ± 2.40	**99.90 ± 0.21**
11	94.38 ± 0.74	93.60 ± 1.77	**94.43 ± 1.12**	86.38 ± 1.79	91.84 ± 1.09	91.19 ± 3.71	85.10 ± 2.25	89.92 ± 1.23	84.49 ± 1.32	84.56 ± 3.86	92.21 ± 2.13
12	99.89 ± 0.21	99.34 ± 0.31	**100.00 ± 0.00**	99.99 ± 0.03	**100.00 ± 0.00**	99.95 ± 0.08	99.47 ± 0.71	**100.00 ± 0.00**	99.45 ± 0.58	99.70 ± 0.35	**100.00 ± 0.00**
13	**99.89 ± 0.17**	99.52 ± 0.41	99.82 ± 0.23	98.37 ± 0.85	99.31 ± 0.33	99.45 ± 0.98	98.62 ± 0.82	99.25 ± 0.47	99.51 ± 0.53	99.71 ± 0.29	99.80 ± 0.28
14	99.06 ± 0.84	**99.85 ± 0.21**	99.55 ± 0.32	99.22 ± 0.59	99.80 ± 0.27	99.30 ± 0.58	98.80 ± 0.63	99.59 ± 0.27	99.01 ± 0.90	98.02 ± 2.04	99.40 ± 0.63
15	94.32 ± 0.95	88.29 ± 1.36	93.08 ± 1.00	91.04 ± 1.32	92.62 ± 1.31	94.94 ± 1.40	90.03 ± 1.39	86.16 ± 1.90	83.28 ± 1.57	86.53 ± 2.43	**96.05 ± 1.91**
16	99.93 ± 0.11	99.99 ± 0.02	**100.00 ± 0.00**	99.98 ± 0.07	99.90 ± 0.28	99.99 ± 0.02	99.77 ± 0.48	**100.00 ± 0.00**	99.91 ± 0.15	98.91 ± 1.21	99.99 ± 0.02
OA	98.26 ± 0.18	96.63 ± 0.25	97.83 ± 0.13	97.25 ± 0.18	97.95 ± 0.29	98.35 ± 0.33	97.08 ± 0.21	95.92 ± 0.32	95.19 ± 0.26	95.55 ± 0.61	**98.72 ± 0.28**
AA	98.52 ± 0.16	97.74 ± 0.13	98.43 ± 0.07	97.38 ± 0.13	98.30 ± 0.24	98.42 ± 0.35	97.40 ± 0.22	97.42 ± 0.16	96.53 ± 0.17	96.40 ± 0.59	**98.82 ± 0.17**
Kappa	98.06 ± 0.20	96.24 ± 0.27	97.59 ± 0.14	96.94 ± 0.19	97.72 ± 0.32	98.17 ± 0.37	96.75 ± 0.24	95.46 ± 0.36	94.65 ± 0.29	95.04 ± 0.68	**98.57 ± 0.32**

TABLE VIII
COMPARATIVE EXPERIMENTAL RESULTS ON THE LK DATASET

Class	SSRN [11]	HybridFormer [33]	EATN [34]	DBMGNet [35]	CVSSN [36]	DBCTNet [37]	FourierFormer [38]	DCTN [39]	SQSFormer [40]	MassFormer [41]	DSGAN
1	98.53 ± 0.75	98.95 ± 0.10	99.44 ± 0.07	98.80 ± 0.17	**99.70 ± 0.12**	99.22 ± 0.29	97.19 ± 0.47	98.03 ± 0.70	98.09 ± 0.51	96.35 ± 0.99	99.69 ± 0.08
2	86.56 ± 6.65	96.16 ± 0.85	71.07 ± 2.26	96.80 ± 1.61	**96.81 ± 1.65**	95.21 ± 4.45	90.76 ± 3.03	90.93 ± 2.25	69.52 ± 3.97	81.86 ± 6.02	90.68 ± 4.63
3	95.51 ± 4.99	96.62 ± 1.73	79.42 ± 3.94	97.37 ± 2.20	96.77 ± 2.39	**99.39 ± 0.50**	92.09 ± 5.69	84.73 ± 8.83	80.24 ± 5.45	82.32 ± 11.04	99.32 ± 0.52
4	98.42 ± 0.50	99.34 ± 0.07	98.82 ± 0.23	99.29 ± 0.15	99.14 ± 0.23	98.89 ± 0.20	**99.49 ± 0.08**	98.72 ± 0.32	98.58 ± 0.23	98.80 ± 0.29	99.18 ± 0.22
5	57.22 ± 7.11	65.05 ± 2.53	77.58 ± 2.70	63.49 ± 4.86	61.63 ± 3.39	63.67 ± 5.94	55.18 ± 3.22	44.48 ± 4.52	50.07 ± 6.11	44.16 ± 5.26	**78.78 ± 9.28**
6	96.82 ± 3.76	99.41 ± 0.27	99.90 ± 0.12	99.59 ± 0.53	98.94 ± 1.00	99.76 ± 0.16	99.26 ± 0.41	97.35 ± 1.09	98.72 ± 0.76	98.44 ± 1.29	**99.90 ± 0.05**
7	**99.98 ± 0.02**	99.86 ± 0.06	99.96 ± 0.01	99.83 ± 0.10	99.95 ± 0.03	99.95 ± 0.02	99.87 ± 0.05	99.89 ± 0.07	99.89 ± 0.06	99.88 ± 0.09	99.97 ± 0.01
8	89.64 ± 2.19	**95.33 ± 1.46**	93.36 ± 1.25	94.05 ± 1.39	94.40 ± 2.26	93.41 ± 1.68	90.63 ± 1.65	77.17 ± 3.01	82.75 ± 2.78	89.35 ± 1.92	94.57 ± 1.74
9	83.19 ± 6.38	**96.52 ± 0.65**	90.49 ± 2.68	93.28 ± 1.58	92.44 ± 4.94	90.23 ± 3.88	67.79 ± 3.24	87.52 ± 3.75	85.38 ± 2.18	75.05 ± 8.26	92.86 ± 1.41
OA	96.12 ± 0.42	98.01 ± 0.10	96.59 ± 0.26	97.78 ± 0.23	97.79 ± 0.23	97.63 ± 0.31	95.70 ± 0.37	94.81 ± 0.61	94.63 ± 0.48	94.58 ± 0.43	**98.31 ± 0.29**
AA	89.54 ± 1.22	94.14 ± 0.27	90.00 ± 0.79	93.61 ± 0.61	93.31 ± 0.56	93.30 ± 0.77	88.03 ± 0.94	86.54 ± 1.63	84.80 ± 1.25	85.13 ± 1.55	**94.99 ± 0.96**
Kappa	94.93 ± 0.55	97.39 ± 0.13	95.54 ± 0.34	97.10 ± 0.30	97.11 ± 0.30	96.90 ± 0.40	94.41 ± 0.47	93.24 ± 0.78	93.00 ± 0.63	92.94 ± 0.55	**97.79 ± 0.38**

TABLE IX
ABLATION EXPERIMENTAL RESULTS ON THE PU AND HOUSTON2013 DATASET

Case	Component							PU			HU2013		
	SRSFE		HDFE		FAGA			OA(%)	AA(%)	Kappa(%)	OA(%)	AA(%)	Kappa(%)
	Sta	CCR	MS	DSN	GT	Conv	Res						
1	×	✓	✓	✓	✓	✓	✓	97.86	97.27	97.16	93.95	94.56	93.46
2	✓	×	✓	✓	✓	✓	✓	98.49	98.07	97.99	94.2	94.61	93.75
3	—	—	✓	✓	✓	✓	✓	98.07	97.80	97.43	90.76	92.90	90.00
4	✓	✓	×	✓	✓	✓	✓	98.89	98.24	98.53	95.95	95.65	95.62
5	✓	✓	✓	×	✓	✓	✓	98.94	98.46	98.59	95.70	94.95	95.35
6	✓	✓	—	—	✓	✓	✓	98.88	98.21	98.51	95.93	95.56	95.60
7	✓	✓	✓	✓	×	✓	✓	98.61	98.14	98.16	95.18	95.16	94.78
8	✓	✓	✓	✓	✓	×	✓	98.95	98.45	98.60	94.70	94.90	94.27
9	✓	✓	✓	✓	✓	✓	×	98.14	97.50	97.53	93.32	93.74	92.78
10	✓	✓	✓	✓	✓	✓	✓	**98.96**	**98.53**	**98.62**	**95.97**	**95.91**	**95.64**

2) *Qualitative analysis:* Figs. 9–12 present the classification results of the proposed SSSGAN method and its competitors across four HSI datasets. It can be seen that the proposed method outperforms other competitors, particularly in classifying complex category boundaries. Comparatively, both MassFormer and SQSFormer exhibit inferior results due to their limited local feature extraction capabilities. The proposed SSSGAN method, however, demonstrates notable improvements in classification accuracy at category boundaries. This is attributed to its hierarchical fusion and dual enhancement mechanisms, which optimize local feature representation and improve the model's sensitivity to subtle variations, allowing it to capture more discriminative features. The classification results across all datasets consistently align more closely with real land category distributions,

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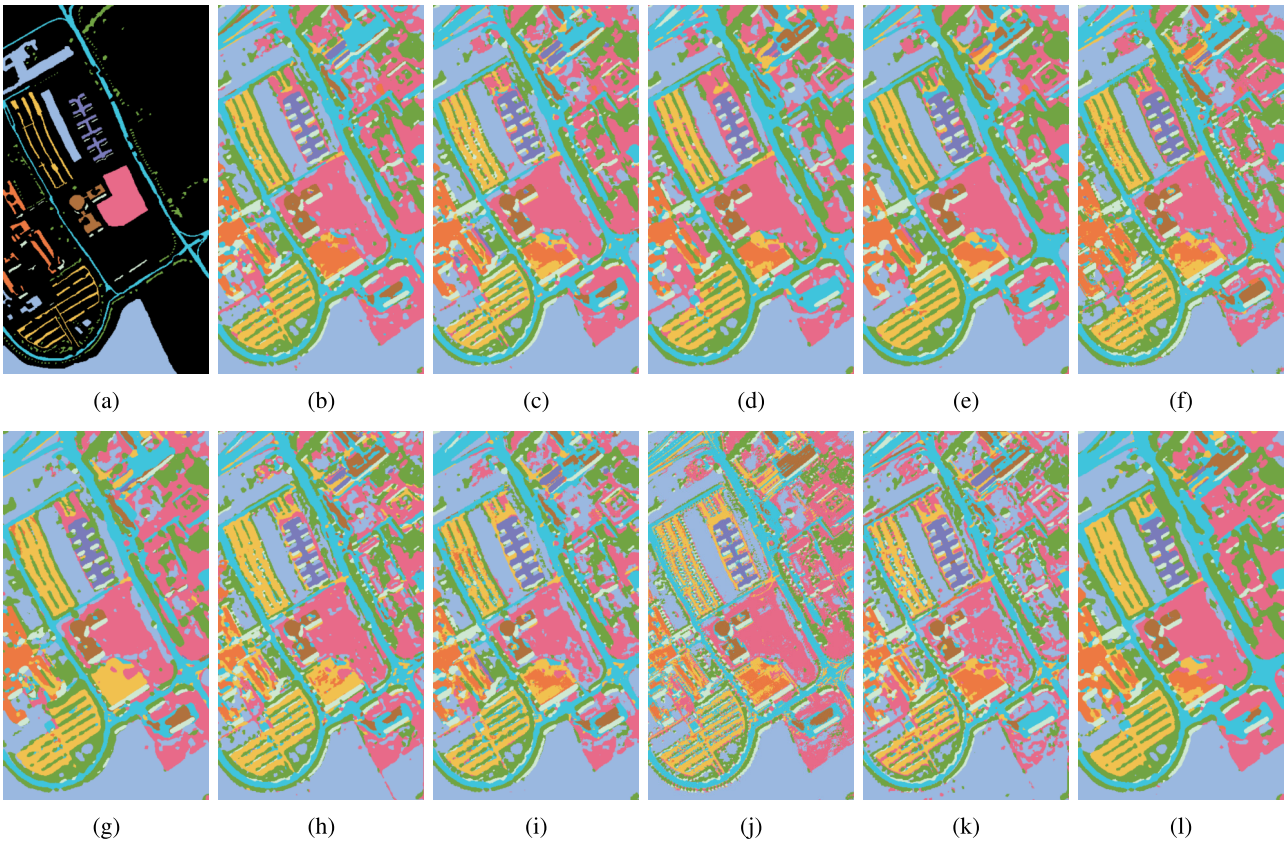


Fig. 9. Visualization of the classification maps using different methods on the PU dataset. (a) Ground truth. (b) SSRN. (c) HybridFormer. (d) EATN. (e) DBMGNet. (f) CVSSN. (g) DBCTNet. (h) FourierFormer. (i) DCTN. (j) SQSFormer. (k) MassFormer. (l) Ours.

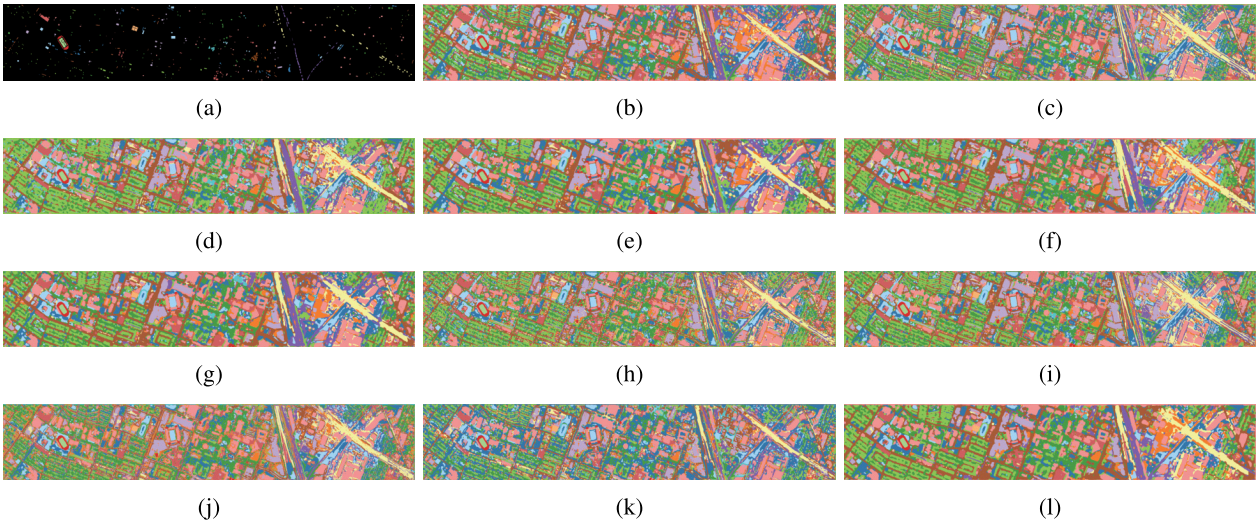


Fig. 10. Visualization of the classification maps using different methods on the HU2013 dataset. (a) Ground truth. (b) SSRN. (c) HybridFormer. (d) EATN. (e) DBMGNet. (f) CVSSN. (g) DBCTNet. (h) FourierFormer. (i) DCTN. (j) SQSFormer. (k) MassFormer. (l) Ours.

demonstrating the superiority of the proposed SSSGAN method in HSIC tasks.

E. Ablation Study

1) Exploration of the effectiveness of primary modules: To verify the effectiveness of each module in our method, we carried out ablation experiments across two datasets to evaluate the three primary modules, that is, SRSFE, HDFE, and FAGA. Therefore, we constructed ten variants to assess the effectiveness of each substructure. Table IX shows the ablation results of the proposed method on the PU and HU2013 datasets. Case 10 serves as the full SSSGAN model in this experiment, acting as a benchmark for comparison with other ablation experiment scenarios.

Cases 1, 2, and 3 are ablation experiments designed to evaluate the proposed SRSFE module. Specifically, in Case 1, the stacked scale-variable convolutional blocks in the SRSFE module were eliminated and substituted with four parallel

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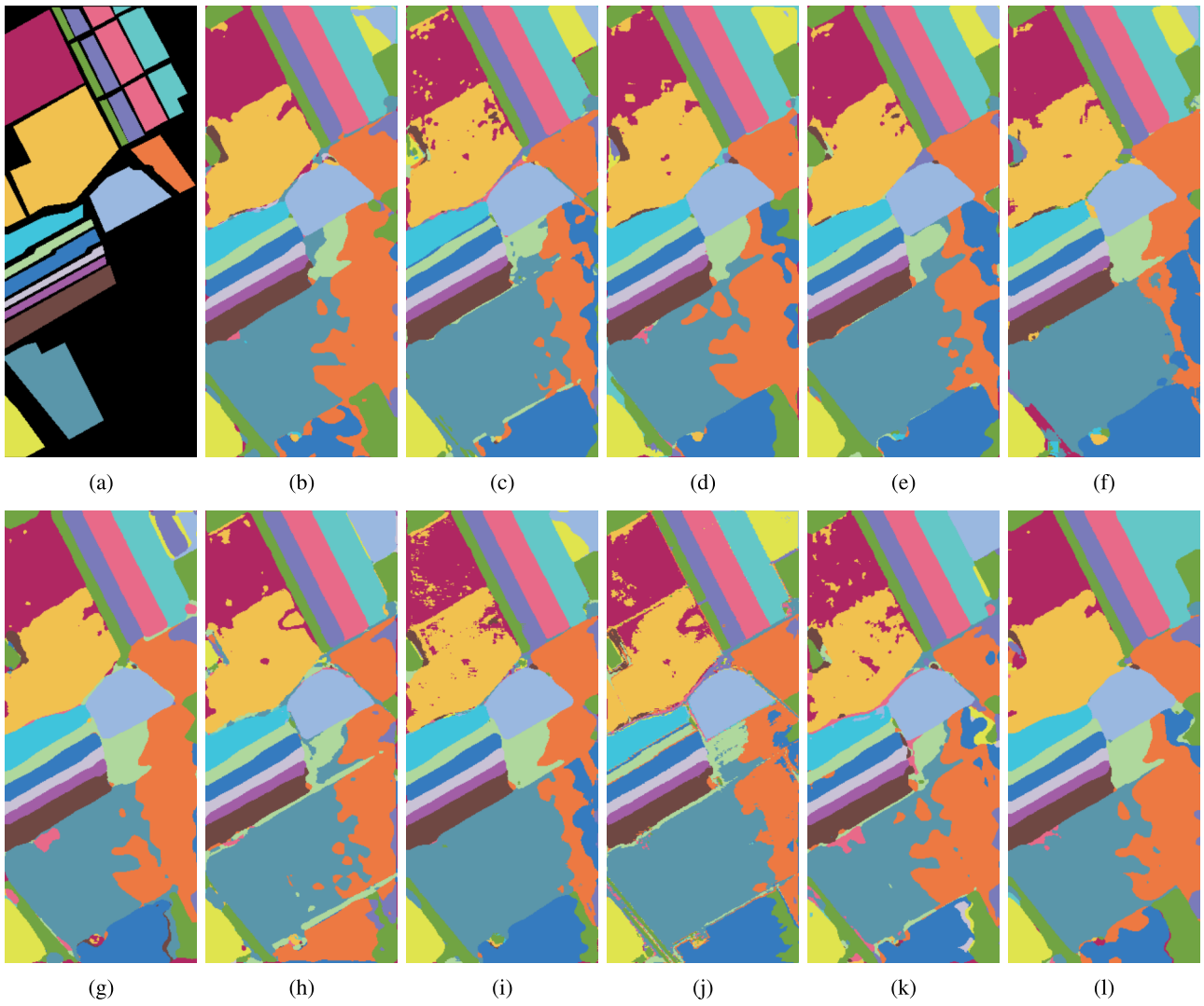


Fig. 11. Visualization of the classification maps using different methods on the SV dataset. (a) Ground truth. (b) SSRN. (c) HybridFormer. (d) EATN. (e) DBMGNet. (f) CVSSN. (g) DBCTNet. (h) FourierFormer. (i) DCTN. (j) SQSFormer. (k) MassFormer. (l) Ours.

multiscale convolutional blocks. In Case 2, the CCR structure in the SRSFE module was replaced with a standard residual structure. In Case 3, the SRSFE module was entirely omitted, retaining only the HDFE and FAGA components. For Cases 1 and 2, Case 1 exhibits a larger accuracy drop compared to our proposed architecture, confirming the superiority of the SRSFE module in local spectral feature extraction. Furthermore, the outcomes of Case 2 underscore the critical role of the CCR structure in HSIC tasks, demonstrating its ability to effectively integrate spectral features across multiple layers while preserving the fine-grained features of the original input.

Cases 4, 5, and 6 are ablation experiment scenarios of the HDFE module. Specifically, Case 4 employed a single dual-shift network branch in the HDFE module to replace the MS for extracting enhanced features. Conversely, Case 5 excluded the DSN from the HDFE module, thereby directly performing matrix multiplication between the raw input features and those extracted via a 3-D convolution. In Case 6, the HDFE module was completely removed, with only the SRSFE and FAGA components retained. It can be seen that all three variants exhibit lower classification performance than the original module, further demonstrating the HDFE module's superiority in extracting local spatial features. For Cases 4 and 5 on the PU dataset, Case 4 demonstrates the poorest performance, with the AA decreasing by 0.29% relative to the original module. This can be attributed to the multibranch structure of the HDFE module, which captures multiscale spatial-enhanced features. By integrating hierarchical features rather than relying on single-scale features, the module can more comprehensively extract detailed information, thus better exploring complex land-cover distributions. For the HU2013 dataset, the removal of the dual shift operation had a greater impact, with the AA dropping by 0.96%. The primary cause is that the dual shift operation enhances feature differences, allowing the model to detect subtle variations effectively. The results of Cases 3 and 6 are inferior to those of the complete SSSGAN network. Specifically, ablating the SRSFE module results in a notable decline in OA, with a 5.21% decrease on the HU2013 dataset. Overall, the outcomes of Cases 3, 6, and 10 demonstrate the synergistic effects between the SRSFE and HDFE modules.

Cases 7, 8, and 9 represent ablation experiments for the FAGA module. In particular, Case 7 eliminated the GT

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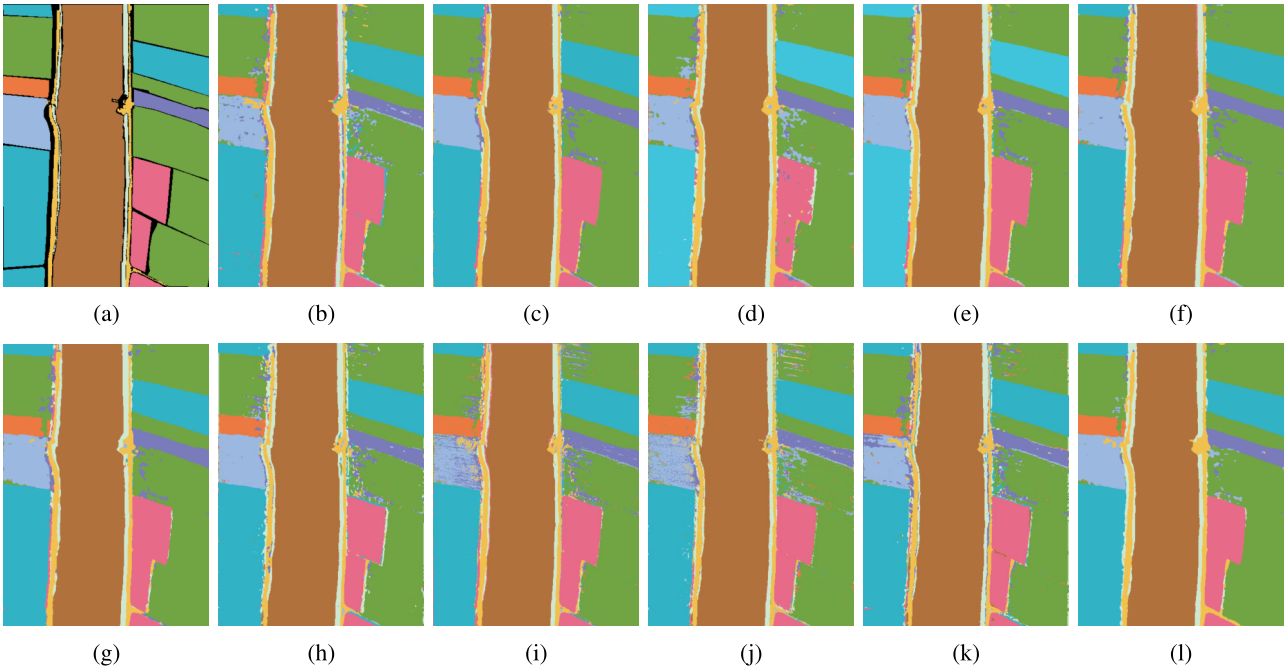


Fig. 12. Visualization of the classification maps using different methods on the LK dataset. (a) Ground truth. (b) SSRN. (c) HybridFormer. (d) EATN. (e) DBMGNet. (f) CVSSN. (g) DBCTNet. (h) FourierFormer. (i) DCTN. (j) SQSFormer. (k) MassFormer. (l) Ours.

TABLE X
COMPUTATIONAL COMPLEXITY COMPARISON OF DIFFERENT MODELS ON THE PU DATASET

Indicators	SSRN	HybridFormer	CVSSN	FourierFormer	DCTN	SQSFormer	vanilla SA	Ours
FLOPs (G)	6.91	2.45	1.28	10.44	40.30	1.43	17.34	12.81
Params (K)	0.40	2.44	0.25	2.04	82.55	0.22	0.11	0.11
Memory (MB)	2394	2626	2576	2488	2964	2554	2558	2530
Infer time (S)	2.37	1.94	0.81	3.29	10.21	0.73	13.73	6.53

Note: FLOPs = Floating Point Operations; Params = Parameters; Memory = Memory Usage; Infer time = inference runtime.

TABLE XI
FSS-SAMPLED HU2013 DATASET: TRAINING AND TESTING ALLOCATIONS

No.	Category	Training	Testing
	Healthy Grass	146	1105
	Stressed Grass	416	838
	Synthetic Grass	91	606
	Trees	170	1074
	Soil	83	1159
	Water	6	319
	Residential	316	952
	Commercial	142	1102
	Road	290	962
	Highway	177	1050
	Railway	159	1076
	Parking Lot1	507	726
	Parking Lot2	44	425
	Tennis Court	79	349
	Running Track	245	415
	Total	2871	12158

in the FAGA module. Case 8 removed the convolution operation within the FAGA module. Case 9 eliminated the global residual operation in the FAGA module. We can see from the ablation experimental results that their classification performances decline in all three scenarios, validating the effectiveness of the FAGA module. On the PU and HU2013 datasets, removing the module's overall residual connection causes a significant decline in classification performance, underscoring the essential role of residual connections in improving model performance. Residual connections can mitigate the vanishing gradient problem and preserve the original feature information, promoting feature reuse and enhancing classification performance.

2) Efficiency Analysis of the Proposed SSSGAN Model: To evaluate the computational efficiency of the proposed model, we conducted a comprehensive comparison of resource consumption with several representative HSIC methods. As presented in Table X, we assessed FLOPs, number of parameters, memory usage, and inference runtime under consistent input conditions. Although the proposed method exhibits higher FLOPs than most alternatives, it significantly reduces the number of parameters. Replacing the guided attention mechanism in the FAGA module with vanilla SA increases FLOPs, inference runtime, and memory usage without altering the number of parameters. This is because the guided token's lower dimensionality effectively reduces FLOPs without introducing additional parameters, thus demonstrating the efficiency of the guided attention mechanism. Despite relatively high FLOPs and inference runtime, the proposed model remains practical, with only 0.11 K parameters and a moderate memory usage of 2530 MB. The increased FLOPs mainly stem from dense matrix operations in the

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TABLE XII
COMPARATIVE EXPERIMENTAL RESULTS ON THE HU2013 DATASET WITH FSS

Class	SSRN	HybridFormer	EATN	DBMGNet	CVSSN	DBCTNet	FourierFormer	DCTN	SQSFormer	MassFormer	DSGAN
1	99.53±0.60	99.07±0.36	98.68 ± 2.47	99.87 ± 0.17	98.22±2.11	99.63±0.80	97.73±1.60	98.91±0.55	**99.93±0.08**	99.89±0.21	99.20±1.44
2	**76.42±1.21**	72.84±0.91	71.98 ± 4.40	75.92 ± 0.63	64.33±2.78	74.84±4.26	69.36±4.54	61.80±1.93	71.38±3.15	75.44±6.66	74.46 ± 4.89
3	99.41±0.85	95.73±5.41	94.33 ± 6.96	92.94 ± 12.40	98.24±3.65	98.32±4.38	94.66±3.84	81.25±6.74	**99.94±0.09**	93.98±2.74	99.92 ± 0.18
4	98.40±0.73	99.08±0.38	95.93 ± 1.37	97.91 ± 0.62	**99.49±0.55**	97.99±1.06	98.78±0.58	99.31±0.38	95.89±1.96	95.30±4.22	98.39 ± 0.66
5	96.05±2.00	89.97±2.18	**96.86 ± 1.69**	91.90 ± 2.81	92.88±1.57	94.49±1.78	96.05±1.37	95.87±0.45	90.88±1.83	95.22±2.08	94.55 ± 2.04
6	4.78±15.12	20.94±33.47	3.64 ± 10.02	4.02 ± 9.56	38.64±49.94	22.50±38.10	10.00±31.62	**48.49±40.25**	7.50±23.72	4.62±11.47	38.75±50.16
7	67.18±3.74	76.20±4.01	66.39 ± 3.89	68.78 ± 1.36	78.31±3.50	68.71±4.10	68.88±4.36	**82.99±2.19**	67.20±3.27	60.68±3.87	68.88±3.04
8	85.13±8.83	94.60±4.88	83.68 ± 6.33	90.72 ± 7.08	83.04±1.95	94.89±7.23	88.74±4.42	84.18±3.03	98.37±1.28	81.87±10.21	**98.42±2.35**
9	77.49±9.59	77.56±4.66	63.30 ± 4.18	**85.59 ± 5.56**	81.47±6.95	73.42±9.17	78.75±3.77	83.94±2.52	77.99±3.97	57.93±6.08	72.74±9.09
10	63.36±3.81	60.35±1.92	**79.61 ± 2.44**	62.12 ± 4.27	63.08±2.99	67.07±6.97	73.07±5.43	70.86±4.93	66.61±3.43	78.90±3.71	73.32±7.94
11	80.28±7.14	96.07±0.72	93.89 ± 2.99	97.00 ± 2.30	89.50±7.00	87.46±7.59	78.68±3.34	79.52±2.23	**97.28±1.42**	85.77±3.43	87.75±3.06
12	47.97±3.51	57.84±2.55	56.41 ± 2.72	**65.29 ± 2.84**	52.60±4.05	57.94±8.14	44.95±2.20	50.94±3.28	51.24±2.62	55.10±5.46	58.86±5.11
13	83.64±8.85	93.01±5.01	**95.43 ± 6.20**	89.97 ± 4.45	82.00±3.89	93.80±4.16	93.66±2.77	91.16±5.40	87.90±3.89	79.45±7.10	91.88±2.99
14	85.88±6.80	93.92±6.08	97.39 ± 7.75	90.47 ± 7.18	**97.67±3.83**	92.37±6.96	84.52±6.51	92.78±3.72	92.86±5.32	81.47±6.79	95.64±4.35
15	**98.38±1.26**	96.53±0.99	78.44 ± 12.78	96.85 ± 0.19	96.02±1.39	95.44±1.28	94.39±2.27	96.65±0.57	97.63±0.58	95.78±1.53	96.94±1.83
OA	78.97±0.90	82.11±0.38	79.23 ± 0.89	82.32 ± 0.52	80.38±0.78	81.30±0.93	78.94±0.98	80.55±0.93	80.58±0.42	77.19±1.56	**82.56±1.10**
AA	77.59±1.27	81.58±2.09	78.40 ± 1.26	80.62 ± 0.60	81.03±3.33	81.26±2.64	78.15±2.54	81.24±2.51	80.17±1.80	76.09±1.61	**83.31±3.82**
Kappa	77.28±0.96	80.65±0.41	77.57 ± 0.95	80.88 ± 0.56	78.79±0.85	79.79±1.00	77.25±1.06	78.98±1.00	79.01±0.45	75.36±1.68	**81.14±1.19**

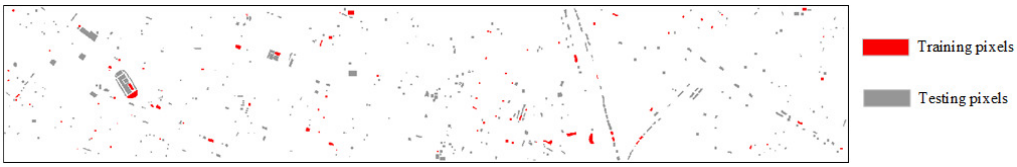


Fig. 13. Results of sampling by FSS on the HU2013 dataset.

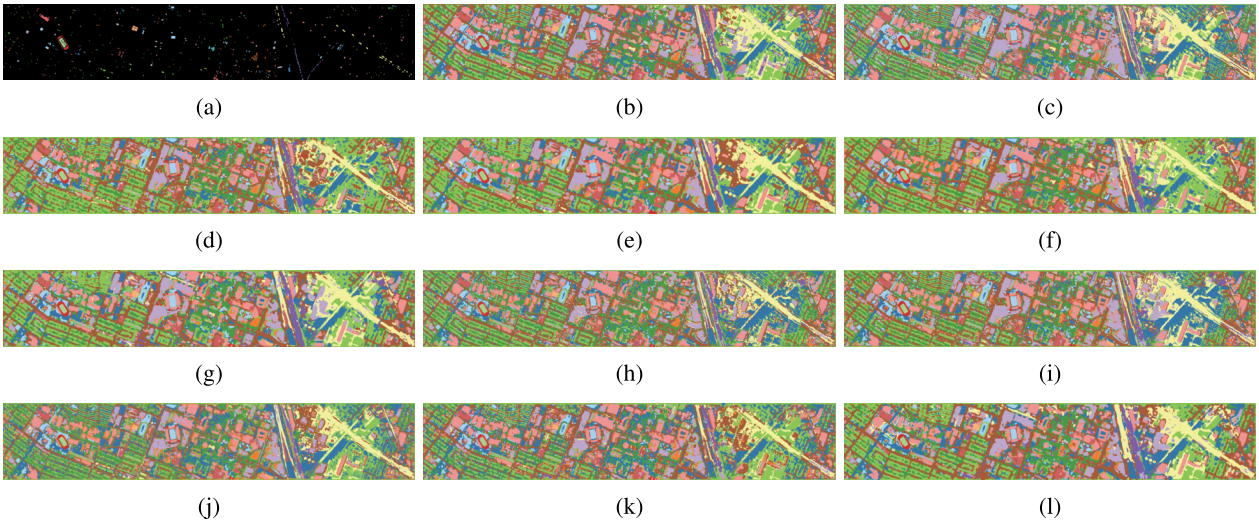


Fig. 14. Visualization of the classification maps using different methods on the HU2013 dataset with FSS. (a) Ground truth. (b) SSRN. (c) HybridFormer. (d) EATN. (e) DBMGNet. (f) CVSSN. (g) DBCTNet. (h) FourierFormer. (i) DCTN. (j) SQSFormer. (k) MassFormer. (l) Ours.

HDFE module, which is designed to extract rich multi-level features. Although this introduces some computational overhead, the guided attention mechanism achieves an effective tradeoff between efficiency and performance. Furthermore, the low number of parameters not only reduces storage requirements but also mitigates overfitting, which is particularly advantageous for HSIC tasks with limited labeled samples.

F. Effectiveness Evaluation of the SSSGAN Method With FSS

In prior experiments, training and testing sets were typically generated through random sampling from the same HSIs. However, this strategy introduces the risk of spatial overlap between the training and testing sets, potentially leading to information leakage. Specifically, image patches centered on neighboring pixels may share similar spatial contexts, which compromises the reliability of model evaluation. To address

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this issue, we adopted the fixed sampling strategy (FSS) [42] to evaluate the effectiveness of the proposed method on the HU2013 dataset.

Fig. 13 illustrates the spatial distribution of the training and testing samples in the HU2013 dataset under the FSS. Table XI presents the number of samples per class for each dataset and Table XII and Fig. 14 present the classification accuracy and visualization results. Although the classification performance of our SSSGAN method slightly decreases under the FSS strategy compared to random sampling, it still outperforms all competitors. Specifically, it achieves the highest OA of 82.56%, AA of 83.31%, and Kappa coefficient of 81.14%. As shown in the classification maps, the proposed SSSGAN method exhibits the clearest class boundaries and superior classification performance. This is primarily attributed to its serially stacked convolutional blocks with varying receptive field sizes, which facilitate fine-grained boundary refinement. Additionally, the dual shift operation enhances interclass separability by amplifying discriminative features. These results further validate the effectiveness and superiority of the proposed SSSGAN method in HSIC tasks.

V. Conclusion

This work proposes an SSSGAN for HSIC. It achieves more accurate boundary refinement and stronger interclass separability by combining an SRSFE with HDFE. Moreover, the proposed FAGA mechanism significantly reduces the computational burden compared with traditional attention modules while preserving powerful discriminative capacity. Therefore, the proposed model effectively balances local spectral–spatial feature extraction and global contextual modeling. Experimental results on four benchmark datasets demonstrate that the proposed method consistently outperforms several state-of-the-art HSIC methods. Due to the significant computational resources required by the proposed method during training, future research could explore dynamic kernel selection strategies to develop a more lightweight architecture while maintaining classification performance. Additionally, the generalization to unseen geographic regions remains a challenge. Thus, a potential direction for future work is to enhance the method's cross-domain adaptability.

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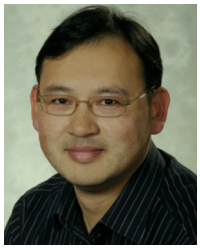
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